

Multiple Choice

1. **Answer:** A

Options: A) by determining the age of lava rising at ocean ridges; B) by determining the composition of limestone deposits; C) by determining the rate of weathering of mountain ranges; D) by determining the type of sedimentary rock formed on land

Note: Scientists arrived at the theory of plate tectonics by determining the age of lava at ocean ridges, which revealed patterns of seafloor spreading and helped establish the movement of tectonic plates over geological time.

2. **Answer:** B

Options: A) Both the caterpillar and the leaf gain energy.; B) Energy is transferred from the leaf to the caterpillar.; C) Decomposers in the leaf obtain energy from the caterpillar.; D) The oak tree gains energy when the caterpillar eats the leaf.

Note: The correct answer is based on the principle of energy transfer in ecosystems, where the caterpillar, as a consumer, obtains energy by consuming the oak leaf, which serves as a source of energy through photosynthesis.

3. **Answer:** C

Options: A) amoeba; B) euglena; C) skin cell; D) bacterial cell

Note: Skin cells must receive nutrients from a transport system because they are part of a larger organism, relying on the circulatory system to deliver essential nutrients and oxygen necessary for their function and survival.

4. **Answer:** B

Options: A) pour the solution down the sink; B) neutralize the solution with a base; C) dilute the solution with more acid; D) absorb the liquid with paper towels

Note: Neutralizing an acid solution with a base is the correct procedure for safe disposal because it minimizes the risk of harmful reactions and transforms the acid into a less hazardous substance, making it safer for the environment and handling.

5. **Answer:** B

Options: A) A mixture; B) An iron; C) A molecule; D) A metal

Note: When a neutral atom gains an electron, it becomes negatively charged and is referred to as an anion, but the answer "an iron" is incorrect; the correct term for a negatively charged ion would be an anion or a specific name based on the element, such as "oxide" for oxygen, not "iron."

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Water vapor is a gaseous form of water that is present in the atmosphere but is not visible to the naked eye, making the statement true.

7. **Answer:** False

Options: A) True; B) False

Note: Heat transferred through the ice cube primarily occurs through conduction, not convection, as convection requires the movement of fluids, which is not applicable in the solid state of an ice cube.

8. **Answer:** True

Options: A) True; B) False

Note: Comparing the skeletons of a recent fish and an ancient fish can reveal structural adaptations and modifications over time that demonstrate the evolutionary processes and relationships among different fish species.

9. **Answer:** True

Options: A) True; B) False

Note: The albumen, or egg white, surrounds the yolk and serves as a protective cushion while also supplying water and nutrients to the developing embryo.

10. **Answer:** False

Options: A) True; B) False

Note: Desert animals are more active at night to avoid the extreme heat of the day and conserve water, not because they absorb sunlight while sleeping.

Long Form Response

11. **Answer:** Euglena and paramecium both have unique reproductive methods that highlight their adaptability, but they differ in their approaches. Euglena primarily reproduces asexually through binary fission, where one organism divides into two identical organisms, allowing for rapid population growth in favorable conditions. In contrast, paramecium can also reproduce asexually through binary fission, but it can engage in sexual reproduction via a process called conjugation, where two paramecia exchange genetic material. This genetic exchange increases genetic diversity, which is important for adapting to changing environments. Both methods are significant in their life cycles, as asexual reproduction allows for quick population increases, while sexual reproduction provides genetic variation, which enhances survival in fluctuating conditions.

Note: correct because it accurately describes the asexual reproductive method of binary fission shared by both Euglena and Paramecium while also highlighting Paramecium's unique capability for sexual reproduction through conjugation, emphasizing the significance of these processes in their adaptability and life cycles.

12. **Answer:** The mixture of salt and water used for cleaning contact lenses is known as saline solution. This combination is effective because salt (sodium chloride) helps to create an isotonic solution, which matches the natural salt concentration of our tears.

This property ensures that the lenses remain comfortable when worn, as it prevents irritation. Additionally, the saline solution helps to rinse away debris and bacteria, keeping the lenses clean and safe for use. Overall, the balance of salt and water in this mixture is essential for maintaining the health of both the lenses and the eyes.

Note: correct because it highlights that saline solution's isotonic nature matches the salt concentration of tears, providing comfort and preventing irritation while effectively cleaning lenses by rinsing away debris and bacteria.

End of Answer Key